

Spectrum-Matched Quantum-Assisted PINNs: A Constructive Design Methodology for Problem-Specific Variational Circuits, Instrumented by Explainable-AI Diagnostics

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September 2026

Abstract

Physics-informed neural networks (PINNs) learn low frequencies fast and high frequencies slowly (spectral bias), because their neural tangent kernel (NTK) spectrum decays with frequency. A data-re-uploading quantum circuit behaves differently: its output is a Fourier series whose frequency set is fixed, exactly, by its encoding. We join the two facts into a constructive design rule, Spectrum-Matched Circuit Design (SMCD): read the frequencies a PDE’s solution needs from the operator’s Fourier symbol and the forcing, then build the circuit whose frequency set covers them, with no architecture search. The hybrid model mixes K copies of the designed circuit through a linear head. Six explainable-AI instruments, run identically on matched classical and hybrid models, test the mechanism rather than only the accuracy, and every claim is pre-registered with a numeric threshold and scored by a script. The circuit produces exactly its designed frequencies (leakage below 4×10^{-15}). Designed frequencies beat random ones of the same size on the heat equation ($6.5\times$, one copy), on Poisson ($236\times$, eight copies) and on a groundwater problem ($9.9\times$, eight copies), on every seed. On waterlogging of farmland from leaking canals, the eight-copy circuit reaches 0.09% error, $18\times$ more accurate than a classical network of the same size, and places the waterlogged stretch to within 1 m. The instruments explain why copies matter: in one circuit every Fourier amplitude depends on the same angles, so the optimiser trades the slow wave for the fast one, and a single copy loses to classical baselines on Poisson. Two results bound the claim: on Poisson a classical model given the same frequencies is $3.3\times$ more accurate than the eight-copy circuit, and at high Helmholtz wavenumbers no model, classical or quantum, learns the solution.

1 Introduction

A classical physics-informed neural network (PINN) learns low-frequency solution components fast and high-frequency components slowly — spectral bias — because its neural tangent kernel (NTK) eigen-spectrum decays with Fourier-mode index. Separately, a data-re-uploading variational quantum circuit realises a truncated Fourier series whose frequency set is determined exactly by the eigenvalues of its data-encoding Hamiltonians [10]. Prior hybrid quantum PINN work [2, 6, 11, 14] reports accuracy on hand-picked circuit sizes, without a principled way to choose the circuit *given* the PDE.

This paper joins the two facts. Given a PDE, the target spectral support is computed from the operator’s Fourier symbol and the forcing and boundary data; the circuit’s encoding scalings, re-uploading depth, qubit count and entangler then follow by formula so that its frequency set covers that support. We call this Spectrum-Matched Circuit Design (SMCD, Section 3). The hybrid model mixes K copies of the designed circuit through a linear head: every copy has the designed frequencies, and the head sets their amplitudes independently.

A design rule is only useful if it can be tested, so the project is built around measurement. Six explainable-AI instruments run at the same checkpoints on matched classical and hybrid models (Section 4), so any difference can be traced to the quantum layer. Every claim about the method was written

down with a numeric threshold and committed to git before the runs that test it, and a script issues every verdict.

Contributions.

1. **SMCD, a constructive map from PDE to circuit** (Section 3): four propositions and an algorithm that output the encoding scalings, depth, qubit count and entangler deterministically, with a design card that can be checked before training.
2. **A hybrid NTK decomposition** for models with disjoint classical and quantum parameters, $\Theta_{\text{hyb}} = \Theta_{\text{cl}} + \Theta_q$, with no cross term (Section 3.5).
3. **Circuit copies**, a model family in which K copies of the designed circuit share its frequency set but have independent amplitudes (Section 3.8).
4. **An explainable-AI protocol for hybrid PINNs**: six instruments with published metric definitions, run identically on every model family (Section 4).
5. **Evidence** from seven model families on six PDE instances and one applied problem, groundwater under irrigated farmland (Sections 5–7).

Findings in brief. The circuit realises exactly the frequencies SMCD designs. Designed frequencies beat random ones of the same size wherever the circuit learns the solution: heat, Poisson and groundwater. One copy is not enough on problems that mix a slow and a fast wave, and the instruments show why; eight copies fix it, reaching 0.09% error on the groundwater problem, 18× better than a classical network of the same size. The method does not beat every classical model: on Poisson a classical model handed the same frequencies stays ahead, on the smoothing heat equation no model beats a plain MLP (as SMCD predicts), and at high Helmholtz wavenumbers every model fails.

2 Background

This section states the results the design rule builds on; full derivations are in the repository’s derivations folder.

2.1 Classical side

PINNs. A PINN [9] minimises the PDE residual at sampled collocation points, with derivatives from automatic differentiation that are exact to floating-point precision. Unlike a Galerkin or finite-element method, the optimisation is non-convex in the network parameters even when the PDE is linear.

The NTK and spectral bias. Writing $\Theta(x, x') = \nabla_{\theta} N[u_{\theta}](x) \cdot \nabla_{\theta} N[u_{\theta}](x')^{\top}$ for the residual’s tangent kernel [5], gradient-flow training in the lazy regime linearises to $\dot{e} = -\Theta_0 e$, so each eigenmode of Θ_0 decays as $\exp(-\lambda_i t)$. For a tanh or ReLU MLP the eigenvalues decay roughly as a power law in Fourier-mode index, so high-frequency components are learned last and slowly (Figure 1). Fourier-feature encodings [12] are the standard classical fix, and therefore a necessary baseline: a quantum layer that beats only a plain MLP has bought nothing Fourier features do not already supply.

Hard boundary conditions. The ansatz $u = B(x) + D(x) N(x)$, with D vanishing on the boundary and B matching the boundary data, satisfies the boundary conditions by construction and removes the boundary-loss block of the NTK.

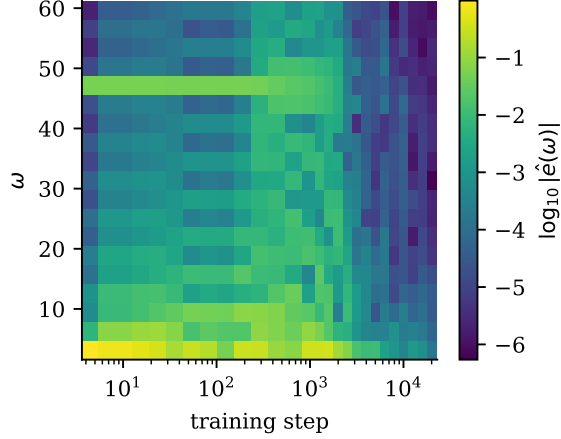


Figure 1: Spectral bias, measured: per-frequency error of a plain MLP over training on the Poisson problem, $u^* = \sin(\pi x) + \alpha \sin(15\pi x)$ with $\alpha = 0.05$. The $\omega = \pi$ mode converges first; $\omega = 15\pi \approx 47$ needs 16.2 times as many steps to reach tolerance (3 seeds).

2.2 Quantum side

Re-uploading circuits are Fourier series. For encoding gates $\exp(-ixH^{(l)})$ with generator eigenvalues $\{\lambda_a^{(l)}\}$, an L -layer re-uploading circuit’s output is a truncated Fourier series whose frequencies are the signed sums of per-layer eigenvalue differences [10]. This is the fact SMCD engineers on (Proposition 1).

Exact derivatives on hardware. The parameter-shift rule $\partial_\theta f = \frac{1}{2}[f(\theta + \pi/2) - f(\theta - \pi/2)]$ holds for Pauli-generated gates, and because the input x enters as a rotation angle, $\partial_x f$ is also exact by a shift. The PINN residual can therefore be computed exactly on quantum hardware, not only in simulation.

Barren plateaus. $\text{Var}[\partial_\theta C]$ decays exponentially in qubit count for a global cost on a deep random circuit [7]. SMCD avoids this by construction: the depth rule picks the *minimum* depth that covers the target band, the observable is local, and angles start small.

Effective dimension. The effective dimension of Abbas et al. [1], from the empirical Fisher spectrum, measures capacity beyond parameter count and is reported for every family.

3 Methodology: Spectrum-Matched Circuit Design (SMCD)

3.1 Setup

The hybrid ansatz used throughout (the *serial* form; Section 5 ablates parallel and random-scaling variants) is

$$x \xrightarrow{E_\varphi} z \in \mathbb{R}^m \xrightarrow{Q_\theta} \phi(z) \in \mathbb{R} \xrightarrow{w} u(x),$$

where E_φ is an affine classical encoder, Q_θ a data-re-uploading variational quantum circuit, and w a linear head. The split is deliberate and is itself a claim: the classical part learns the coordinate *warping*, the quantum part supplies the Fourier *basis*, and the head learns the *coefficients* — exactly the structure of a spectral method with a learned coordinate map.

3.2 Proposition 1 — Circuit spectrum

Proposition 1 (Circuit spectrum). *An L -layer data-re-uploading circuit, with encoding gate $\exp(-ix H^{(l)})$ at layer l for Hermitian generator $H^{(l)} = \sum_a \lambda_a^{(l)} |a\rangle\langle a|$, realizes an observable expectation of the form*

$$f(x) = \sum_{\omega \in \Omega} c_{\omega}(\theta) e^{i\omega x}, \quad \Omega = \left\{ \sum_{l=1}^L \left(\lambda_{a_l}^{(l)} - \lambda_{b_l}^{(l)} \right) \right\},$$

i.e. a trigonometric polynomial whose frequency support is the set of all signed differences of encoding-generator eigenvalues, summed over layers.

Proof sketch. Expand $f(x) = \langle 0|U^\dagger(x, \theta) M U(x, \theta)|0\rangle$ by inserting the eigenbasis resolution of each diagonal encoding gate independently on the bra and ket sides. Every (bra-path, ket-path) pair contributes an x -dependent phase $\exp(ix \sum_l (\lambda_{a_l}^{(l)} - \lambda_{b_l}^{(l)}))$ multiplied by an x -independent amplitude collecting the trainable gates and M along that path; collecting paths by total frequency gives the boxed form. The full derivation, worked first for one qubit/one layer and then generalized to n qubits and d -dimensional inputs, is in docs/derivations/07_circuit_fourier_spectrum.md. ■

Two subtleties that change the constructive guarantee. Two facts, easy to get wrong and verified numerically against both an independent statevector simulator and a PennyLane oracle (agreement $< 10^{-10}$) before being trusted:

1. **Generator-vs-frequency factor of two.** The generator’s eigenvalues are $\pm\omega/2$ for an $RZ(\omega x)$ gate, but the *realized* frequency in $f(x)$ is the full ω , not $\omega/2$: the eigenvalue-difference structure above supplies a compensating factor of two. Reading the generator’s eigenvalues off the gate and reporting that as the frequency halves every number in the design.
2. **State-preparation degeneracy.** If the qubit is prepared exactly at the equator ($RY(\pi/2)$ before the first encoding gate, an equal superposition), every frequency whose balanced-ternary leading digit is zero — exactly one third of Ω — has identically zero amplitude for *every* trainable-parameter setting, independent of ansatz expressivity. This is not a numerical inconvenience: it makes Proposition 3 below false for any target spectrum touching a dead frequency under the literal $\pi/2$ prescription. We use $\phi_{\text{prep}} = \pi/3$ throughout, verified to have no dead frequencies.

3.2.1 Corollary: ternary scaling and the depth rule

Set the per-layer scalings to $\omega_l = \Delta \cdot 3^{l-1}$ (single-wire case). Then $\Omega = \Delta \cdot \{-\frac{3^L-1}{2}, \dots, \frac{3^L-1}{2}\}$ — exactly 3^L distinct frequencies with no degeneracy, a bijection with the balanced-ternary representation of the integers in that range (proof: substitute $m_l = d_l - 1$ to convert the signed sum into an ordinary base-3 expansion shifted by $\frac{3^L-1}{2}$). This grows *exponentially* in depth, versus $2L + 1$ distinct frequencies for linear scaling $\omega_l = \omega_0$ — the design knob almost every prior re-uploading paper leaves fixed.

To guarantee the reachable band covers a target maximum frequency K at ternary spacing Δ :

$$\max |\Omega| \geq K \iff \Delta \frac{3^L-1}{2} \geq K \iff L \geq \log_3 \left(\frac{2K}{\Delta} + 1 \right) \implies \boxed{L = \lceil \log_3 \left(\frac{2K}{\Delta} + 1 \right) \rceil}.$$

Both the factor of two inside the logarithm and the $+1$ matter. For the Poisson target spectrum ($K = 15\pi$, $\Delta = \pi$), the naive $L = \lceil \log_3(K/\Delta) \rceil$ gives $L = 3$, whose reachable band tops out at $13\pi < 15\pi$ and silently misses the target mode; the rule above gives $L = 4$, whose band reaches $40\pi \geq 15\pi$.

3.3 Proposition 2 — PDE target spectrum

Proposition 2 (PDE target spectrum). *For a linear constant-coefficient operator $\mathcal{L}$ with Fourier symbol $\sigma(k)$, the solution of $\mathcal{L}u = f$ on a periodic/rectangular domain satisfies $\hat{u}(k) = \hat{f}(k)/\sigma(k)$. The essential spectral support $S_\varepsilon = \{k : |\hat{u}(k)| > \varepsilon \|\hat{u}\|\}$ is therefore computable before training, from $\hat{f}$ and σ alone.*

For Helmholtz, $\sigma(k) = k^2 - |\mathbf{k}|^2$, so S_ε concentrates near $|\mathbf{k}| \approx k$ — the near-resonant band a classical PINN’s low-frequency-first learning dynamics reach last, if at all. For the heat equation, σ grows quadratically with $|\mathbf{k}|$ and the solution decays exponentially in time at every nonzero mode, so S_ε collapses toward $k = 0$ as t increases: this is the project’s negative control (Section 5.2) — there is no high-frequency content for a quantum layer to supply, and SMCD predicts zero benefit *a priori*, not just observes it after the fact.

3.4 Proposition 3 — Matching condition and resource bound

Proposition 3 (Matching & resource bound). *If $S_\varepsilon \subseteq \Omega$, the hybrid model represents the solution to ε accuracy with a linear head — the remaining optimization problem is convex in w . Covering S_ε with maximum frequency K and resolution Δ requires $L \geq \log_3(2K/\Delta + 1)$ layers (ternary scaling, above) and $n \geq \lceil d \rceil$ qubits for d input dimensions, plus ancilla wires for any required cross-frequency terms.*

This turns architecture choice into arithmetic: given (K, Δ, d) read off the PDE’s own symbol and data, (L, n) follow by formula, not by search.

3.5 Proposition 4 — NTK consequence

Proposition 4 (Hybrid NTK decomposition). *For a hybrid model whose parameters $\theta = (\theta_{\text{cl}}, \theta_q)$ partition into disjoint classical and quantum groups,*

$$\Theta_{\text{hyb}}(x, x') = \Theta_{\text{cl}}(x, x') + \Theta_q(x, x'),$$

*exactly, with **no cross term**; a term of the form $2\Theta_\times$ does not arise for a shared-output, disjoint-parameter model.*

Proof. $\Theta(x, x') = \sum_{k=1}^P \partial_{\theta_k} f(x) \partial_{\theta_k} f(x')$ is a single inner product of gradient *vectors*, summed over the full index set of parameters $k = 1, \dots, P$. When that index set is partitioned into two disjoint blocks θ_{cl} and θ_q (a hybrid model’s classical and quantum parameters never overlap), the sum splits additively over the partition:

$$\Theta_{\text{hyb}}(x, x') = \sum_{k \in \text{cl}} \partial_{\theta_k} f(x) \partial_{\theta_k} f(x') + \sum_{k \in q} \partial_{\theta_k} f(x) \partial_{\theta_k} f(x') = \Theta_{\text{cl}}(x, x') + \Theta_q(x, x').$$

There is no third term: a cross term would only arise from a fundamentally different quantity, such as a covariance between two *separately* defined kernels or the square of a sum of outputs — not from the NTK of a single model with disjoint parameter groups, which this project’s ansatz is. The identity is also checked numerically on random synthetic Jacobians of matched and mismatched sizes; `src/qapinn/xai/ntk.py` implements it. ■

Because Ω (the quantum block’s realized frequency set) is discrete and prescribed by construction, Θ_q has eigen-directions *at* the encoded frequencies with eigenvalues bounded below independently of $|n|$ for $n \in \Omega$ — the exponential/power-law eigenvalue decay in $|k|$ that drives classical spectral bias (Section 2) is replaced by a flat response over Ω . This is the formal statement of “the quantum layer removes spectral bias inside its band,” and it is what Section 5’s NTK spectrum figures measure directly: decay exponent inside Ω versus outside it, for matched classical and hybrid models.

Algorithm 1 Spectrum-Matched Circuit Design (SMCD)

Require: PDE operator $\mathcal{L}$, forcing f , boundary data g , domain Ω_x , tolerance ε , qubit budget $n_{\max}$

Ensure: encoding generators & scalings $\{\omega\}$, depth L , qubits n , entangler, observable, ansatz form

- 1: **Symbol:** $\sigma(k) \leftarrow$ Fourier symbol of $\mathcal{L}$ (linearize about a base state if nonlinear; Burgers uses the viscous-scale estimate $k_{\max} \sim 1/\sqrt{\nu}$)
 - 2: **Target:** $\hat{S} \leftarrow \{k : |\hat{f}(k)/\sigma(k)| > \varepsilon\} \cup$ boundary-lift spectral support (time-dependent: union over t , record growth rate)
 - 3: **Band:** $K \leftarrow \max |k|$ in $\hat{S}$; $\Delta \leftarrow$ min spacing in $\hat{S}$; $d \leftarrow \dim(x)$
 - 4: **Depth:** $L \leftarrow \lceil \log_3(2K/\Delta + 1) \rceil$ with ternary scalings $\omega_j = \Delta \cdot 3^{j-1}$ (Section 3.2; fall back to linear scaling if $\hat{S}$ is a dense low band)
 - 5: **Width:** $n \leftarrow d + n_{\text{cross}}$, where n_{cross} covers required mixed-frequency terms (e.g. 2-D Helmholtz needs $k_x \pm k_y \Rightarrow$ entangle the two coordinate wires)
 - 6: **if** $n > n_{\max}$ or $L > L_{\max}$ (barren-plateau depth budget) **then**
 - 7: split the band into an octave ensemble (Section 3.7)
 - 8: **end if**
 - 9: **Entangler:** ring-CZ if cross terms are needed, else none (unnecessary entanglement costs gradient variance for no benefit on a separable target)
 - 10: **Observable:** local $\langle Z_0 \rangle$ (never a global Pauli string — barren plateaus)
 - 11: **Ansatz:** hard-constrained BCs, $u = B(x) + D(x) \cdot [w \cdot Q_\theta(E_\varphi(x))]$
 - 12: **Init:** small-angle $\theta \sim \mathcal{N}(0, \sigma_0^2)$, $\sigma_0 = 0.1$; identity-block init for the encoder
 - 13: **Report:** emit a design card ($\hat{S}, \Omega$, coverage $= |\hat{S} \cap \Omega|/|\hat{S}|$, L, n , param count, predicted NTK band), committed alongside every experiment run
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3.6 Algorithm 1 — SMCD

The design card (final step) is what makes the methodology auditable: every run’s predicted coverage can be set against its measured error (Section 5.4).

3.7 Octave-split circuit ensembles

When a single circuit cannot cover $\hat{S}$ within the barren-plateau depth budget ($L > L_{\max}$), SMCD falls back to M shallow circuits, each matched to one octave band $[2^j \Delta, 2^{j+1} \Delta)$ of $\hat{S}$, summed at the head. This keeps every individual circuit shallow (trainable) while the ensemble covers a wide band — multi-resolution analysis with a quantum basis per resolution level.

3.8 Circuit copies

A single circuit reads all of its Fourier amplitudes out through one expectation value, so every amplitude is a function of the same few angles: changing one frequency’s amplitude moves the others. The copy model removes that coupling without changing the frequency set. It evaluates K copies of the SMCD circuit on the same encoded input, each with its own trainable angles, and mixes them with a linear head,

$$u(x) = B(x) + D(x) \left[b + \sum_{j=1}^K w_j Q_{\theta_j}(E_\varphi(x)) \right].$$

Every copy has the same encoding scalings, so the model’s frequency set is still exactly Ω and every design-card statement above applies unchanged; only the number of independent handles on the amplitudes grows. $K = 1$ is the plain SMCD circuit. On the one-dimensional problems each copy is a one-qubit, four-layer circuit, and $K = 8$ gives 91 trainable parameters. K and the learning rate were selected on tuning seeds before any held-out run (Section 4.2).

4 Measurement protocol

4.1 Explainable-AI instruments

Six instruments run at the same checkpoints (0, 100, 500, 1k, 5k, 20k steps and the final step) for every model, problem and seed, so a measured difference cannot come from when or how each family was probed (Table 1). Each instrument’s expected signature is stated as a pre-registered claim, which guards against the post-hoc storytelling PINN explainability work is criticised for [8].

Instrument	What it measures	Expected signature for SMCD
NTK spectroscopy [3]	eigenvalues, decay exponent, effective rank and drift $\ \Theta_t - \Theta_0\ _F / \ \Theta_0\ _F$ of the empirical NTK	flatter spectrum inside Ω than outside (Proposition 4)
Per-frequency error	error $u_\theta - u^*$ projected on Fourier modes at every checkpoint	improvement over the MLP concentrated inside Ω
Attribution	Integrated Gradients of the residual over the domain, with completeness violation and correlation to the true residual	attributions track the residual
Fisher information	empirical Fisher spectrum and effective dimension [1]	separates expressivity from parameter count
Gradient variance	$\text{Var}[\partial_\theta L]$ against qubit count and depth	stays above the 2^{-n} barren-plateau line [7]
Probes and landscapes	per-layer ridge-regression R^2 ; 2-D loss slices around the trained point	trained point is a minimum

Table 1: The six instruments, run identically on classical and hybrid models.

4.2 Experimental setup

Problems. Six one- and two-dimensional instances on the unit domain, all with exact or high-accuracy references: Poisson, $-u'' = f$ with solution $\sin \pi x + 0.3 \sin 15\pi x$, a slow and a fast wave; the heat equation with a 15π ripple in the initial condition, a smoothing operator; viscous Burgers, nonlinear; and two-dimensional Helmholtz at $k = 4, 10, 20$. A seventh, groundwater flow under irrigated farmland, is the application of Section 7.

Model families. Every family uses the same hard boundary ansatz and training loop. The SMCD circuit (one copy, or K copies, Section 3.8); the same circuit with random encoding scalings drawn from $U(0.5, 3)$, the size-matched test of the design itself; a parallel hybrid (an MLP beside the circuit) and an octave-split ensemble (Section 3.7); an MLP (8,513 parameters on Poisson), a Fourier-feature network (129) and a frequency-matched classical model, a linear head on sine and cosine features at exactly the frequencies SMCD targets, sized to match the circuit. SMCD builds one-qubit, four-layer circuits for Poisson, heat and Burgers ($|\Omega| = 81$, 14–18 parameters) and three-qubit, one-layer circuits with a ring of CZ gates for Helmholtz ($|\Omega| = 15$). Only the frequency-matched model and the random-frequency circuit are size-matched to the circuit; the MLP and Fourier-feature network run at their standard sizes. On the groundwater problem an MLP of the circuit’s size is added.

Training. 20,000 Adam steps followed by 2,000 L-BFGS steps, five seeds per configuration. The one-copy circuit and all six problems use seeds 0–4 at a common learning rate. The eight-copy circuit was selected on the Poisson problem with seeds 0–2 from eight configurations ($K \in \{1, 2, 4, 8\}$, two learning rates), while each classical family received eight learning-rate candidates, under a selection rule fixed before any tuning run; all of its reported results are on held-out seeds 10–14, which played no part in

selection. Circuits are simulated exactly by a statevector simulator cross-checked against PennyLane to 10^{-10} .

4.3 Pre-registered claims and how they are scored

Every claim about the method was written with a deciding metric and a numeric threshold in the repository’s `docs/` folder and committed to git before the runs that test it; a verification script checks that the committing revision is an ancestor of every such run. Claims about the eight-copy circuit were written after model selection on the tuning seeds and before any held-out run. An adjudication script issues every verdict from the committed run records. A claim is *confirmed* when its threshold is met and, for a comparison between two models, all five paired seeds agree (paired Wilcoxon $p \leq 0.0625$, the smallest value five pairs allow); *refuted* when the threshold is missed; *inconclusive* when the median meets the threshold but the seeds disagree. No threshold is changed after a run.

4.4 Implementation

Everything above is implemented in one open codebase (about 5,400 lines of library code, 4,500 of experiment and analysis scripts and 6,300 of tests). Its parts are:

- **A statevector simulator** for data-re-uploading circuits with exact derivatives in the input and the angles, cross-checked against PennyLane to 10^{-10} and against a density-matrix simulation for the noise surrogates.
- **SMCD itself**: the symbol-based target spectrum, the depth and width rules, the octave split and the design card, with a test that every designed circuit emits nothing outside Ω .
- **One training harness** for all seven families (hard or soft boundary conditions, Adam then L-BFGS, deterministic seeding) that records the six instruments at fixed checkpoints and stores every run under a content hash of its configuration.
- **An adjudication pipeline** that reads only the stored run records, issues every verdict, verifies that each claim was committed before its runs, and redraws every figure and table without retraining.

The experiments comprise 327 recorded training runs, about 130 hours of training on one laptop GPU. 402 automated tests and a continuous-integration job rerun the test suite, re-derive every verdict and compare every figure against the committed record on each change.

5 Results

Errors are relative L^2 against the reference solution, reported as the median over five seeds. A relative error of 1 is what the trivial prediction $u \equiv 0$ scores, so a model near 1 has learned nothing, and a model above 1 is worse than predicting zero. Table 2 lists every pre-registered claim about the one-copy circuit and Table 4 those about the eight-copy circuit; the subsections take the evidence one question at a time and say what each result means for the method.

5.1 The circuit builds exactly what SMCD designs

For every one-dimensional problem the SMCD circuit was evaluated at 40 random parameter settings and its output Fourier-transformed. The largest amplitude outside the designed set Ω , relative to the largest overall, is at most 3.8×10^{-15} , which is floating-point zero, and the test suite checks the same property on every build. The frequency set could still drift during training, because the encoder in front of the circuit is trainable; it does not. The encoder drift $\|A - I\|_F$ at the end of training is 0.13 with one copy and 0.10 with eight, below the 0.2 at which the design card would stop describing the trained model.

Claim	Verdict	Measured
Beats the large MLP on Poisson ($\geq 2\times$)	Refuted	$0.65\times$
Beats Fourier features on Poisson ($\geq 1.3\times$)	Refuted	$0.13\times$
Matches the frequency-matched classical model on Poisson (within $1.3\times$)	Refuted	$5.11\times$
Heat: no model beats the MLP by more than 5%	Confirmed	
Helmholtz: the gain over the MLP grows with k	Confirmed	
Designed beats random frequencies, Poisson ($\geq 1.5\times$)	Inconclusive	$1.84\times$
Designed beats random frequencies, heat ($\geq 1.5\times$)	Confirmed	$6.52\times$
NTK flatter inside Ω than outside, Poisson	Insufficient data	
NTK flatter inside Ω than outside, Helmholtz $k=10$	Refuted	gap -8.4
Gain over the MLP lies inside Ω , Poisson ($\geq 70\%$)	Confirmed	99.9%
Gain over the MLP lies inside Ω , Helmholtz $k=10$	Refuted	
Higher coverage gives lower error ($\rho \leq -0.7$)	Refuted	
Gradient variance stays above the barren-plateau line	Confirmed	slope 0.31, R^2 0.001
Leaves the lazy-training regime earlier than classical models	Refuted	1.01 vs $4.3e+06$
Encoder drift stays below 0.2, so Ω stays valid	Confirmed	0.13
Confirmed	6 of 15	

Table 2: Pre-registered claims about the one-copy SMCD circuit, seeds 0–4, as the adjudication script scores them. Ratios are baseline error over circuit error, except the frequency-matched comparison (circuit over baseline) and drift (hybrid vs classical).

What this means. Proposition 1 and the ternary depth rule are not only theory: the design card computed before training is a correct, checkable description of the model after training. Every later statement about “inside” and “outside” Ω rests on this.

5.2 Accuracy across six problems

Table 3 gives every family’s error on every problem with one copy of the circuit. The six problems fall into three regimes.

Family	Poisson	Heat	Burgers	Helmholtz		
				$k=4$	$k=10$	$k=20$
MLP	1.08	0.305	0.014	5.0×10^{-5}	1.00	1.00
Fourier features	0.21	0.39	0.33	0.92	1.00	1.00
Frequency-matched	0.33	0.33	0.31	0.95	1.00	1.00
SMCD circuit	1.68	0.42	1.14	1.59	1.09	1.00
Random frequencies	3.08	2.74	1.06	0.95	1.01	1.01
Parallel hybrid	0.35	0.76	0.26	0.26	1.00	1.00
Octave ensemble	0.75	0.68	0.85	1.00	1.00	1.00

Table 3: Median relative L^2 error, one copy of each circuit, seeds 0–4 (two to four seeds for six Helmholtz $k=10/20$ cells). Bold: best per problem.

Mixed slow and fast waves (Poisson, Burgers, Helmholtz $k=4$): classical models win against one copy. On Poisson the circuit’s error (1.68) is higher than the MLP’s (1.08), the Fourier-feature network’s (0.21) and the frequency-matched model’s (0.33), and all three pre-registered comparisons are refuted. Two details sharpen the reading. The frequency-matched model has the circuit’s exact frequencies and 13 parameters to its 14, so the gap ($5.1\times$) is not about which frequencies are available but about how the model can combine them. And the parallel hybrid, which adds an MLP beside the circuit, reaches 0.35, close to the frequency-matched model: once the model has a second, independent path to

the output, it trains well. Both are consistent with the readout of a single circuit being the bottleneck, which Section 5.5 examines and removes.

Smoothing operator (heat): nothing beats a plain MLP, as predicted. Diffusion damps each Fourier mode at a rate proportional to ω^2 , so the 15π ripple in the initial condition vanishes almost immediately and the solution is dominated by a slow decay that any smooth network represents well. SMCD’s design card predicts a benefit of only 2.5% here before any training. Every family is worse than the MLP, the circuit by 38% and the random-frequency circuit by $9\times$, so the negative-control claim (no family beats the MLP by more than 5%) is confirmed. The practical rule is that SMCD tells a user in advance when a quantum layer is not worth adding.

High wavenumber (Helmholtz $k=10, 20$): every model fails. Every family’s median error is 0.995–1.09, the score of predicting zero. At these wavenumbers the forcing term is about 1% of the Laplacian term, so $u \equiv 0$ already leaves a small residual and gradient descent settles there; this reading is plausible but untested. The pre-registered trend that the circuit’s error ratio against the MLP improves with k holds ($3 \times 10^{-5} \rightarrow 0.92 \rightarrow 0.99$), and the circuit is within $1.3\times$ of the frequency-matched model at both wavenumbers, but these are parities among models that did not learn the solution and say nothing about capability.

Cost. On a classical simulator a circuit run takes a median of about 2,900 s against 330–590 s for the classical families at the same number of steps. This is the price of simulation, not of the method: on hardware a one-qubit circuit evaluation is a single short pulse sequence.

5.3 Designed frequencies beat random ones

The sharpest test of SMCD changes only one thing: the circuit, its size and its training stay fixed, and the encoding scalings are either SMCD’s or random draws from $U(0.5, 3)$ (Figure 2). A worked example shows what is at stake. With four layers, random scalings of at most 3 reach frequencies of at most $4 \times 3 = 12$ radians per unit, while the Poisson and heat solutions contain a $15\pi \approx 47$ wave; SMCD’s scalings $\pi \cdot \{1, 3, 9, 27\}$ reach $40\pi \approx 126$. The random circuit cannot represent the fast wave at any parameter setting. Wherever the circuit learns the solution, the design wins on every seed:

- **Heat, one copy:** $6.5\times$ lower error on all five seeds, 95% bootstrap interval 2.5–17.7 $\times$. The designed circuit is also the most consistent model in the whole matrix (its worst and best seeds differ by 4%), while its random twin is the least consistent (7.4 $\times$).
- **Poisson, eight copies:** $236\times$ (0.0037 against 0.87), all five held-out seeds.
- **Groundwater, eight copies:** $9.9\times$ (0.09% against 0.92%), all five held-out seeds (Section 7).

On Poisson with one copy the median ratio is $1.84\times$ but one seed of five disagrees, so that claim is inconclusive; there the one-copy circuit does not learn the solution with either scaling.

What this means. This is the central claim of the method, and it holds: choosing the encoding from the PDE’s symbol is worth between one and two orders of magnitude in accuracy over the same circuit with an arbitrary encoding, at no extra cost in parameters.

5.4 The mechanism in the frequency domain

The improvement lands on the designed frequencies. Projecting the error onto Fourier modes at every checkpoint and accumulating how much better the circuit does than the MLP at each frequency, 99.9% of the circuit’s total improvement on Poisson lies inside Ω with one copy, 99.9% with eight, and



Figure 2: One copy of the circuit with SMCD’s scalings against the same circuit with random scalings, every seed: heat (left) and Poisson (right).

99.4% on groundwater; the pre-registered bar was 70%. Figure 3 shows it directly: the MLP’s error at high $|\omega|$ persists through training (the spectral-bias staircase), while the circuit clears the rows marked by Ω . On Helmholtz $k=10$ the circuit improves on the MLP at no frequency at all, so there is nothing to attribute.

What this means. The circuit’s accuracy gains come from exactly the mechanism SMCD is built on, not from an incidental property such as its size or initialisation.

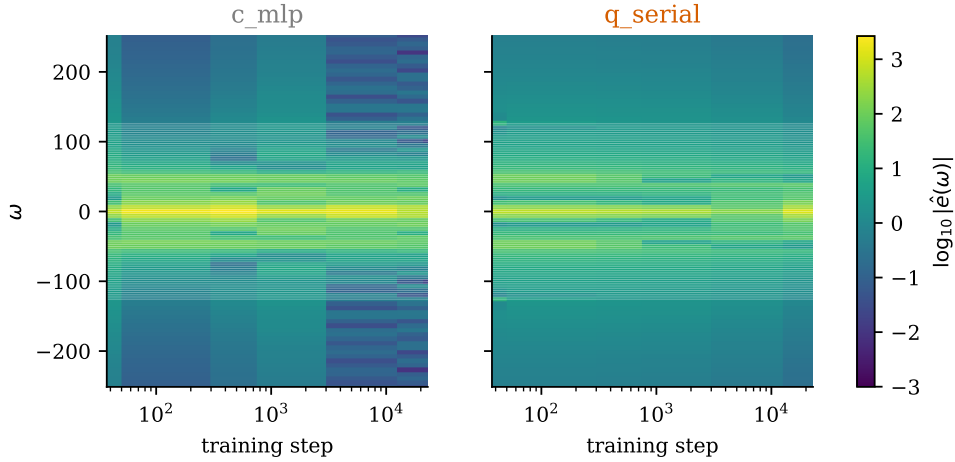


Figure 3: Poisson: per-frequency error $|\hat{e}(\omega)|$ over training, MLP (left, `c_mlp`) against one copy of the SMCD circuit (right, `q_serial`), seed 0, with Ω overlaid as thin white lines.

The NTK is flatter inside Ω once the model can express it. Proposition 4 says the quantum block’s kernel has non-decaying eigen-directions at the encoded frequencies, so its spectrum should be flatter inside Ω than outside. With eight copies on Poisson the decay exponent inside Ω is flatter than outside by 15.2 (bar: 0.5). With one copy the test cannot run on Poisson: the model has 14 parameters, so its NTK has at most 14 non-zero eigenvalues, none beyond index $|\Omega| = 81$ (Figure 4, left). On Helmholtz $k=10$, where every model fails, the spectrum is steeper inside Ω (gap -8.4). The prediction therefore holds exactly where the model has enough independent directions to show it.

Hybrid models stay close to their initial kernel. The prediction that hybrids leave the lazy-training regime earlier is reversed: at 5,000 steps the median relative NTK drift is 1.0 for the one-copy hybrid and 4.3×10^6 for the classical models (2.5 against 9.7×10^4 with eight copies). The circuits also use few directions in parameter space throughout: NTK effective rank 1.4 on Poisson and 1.1 on heat, Fisher effective dimension about 4, both flat over training, whereas the MLP’s effective dimension grows

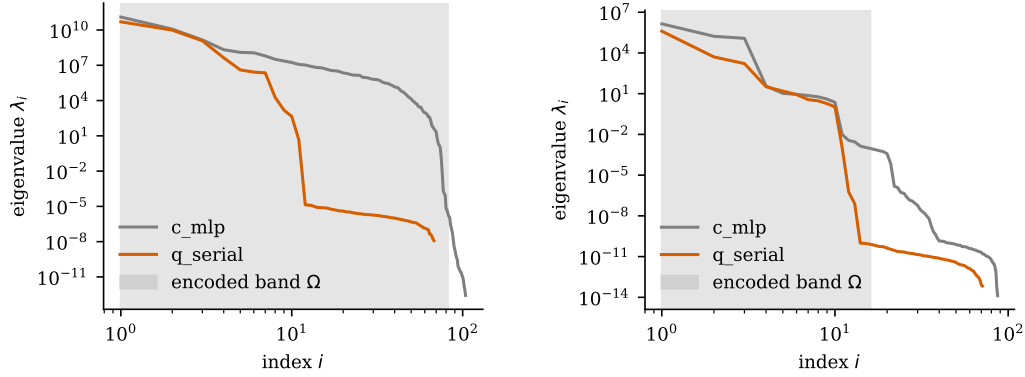


Figure 4: Residual NTK eigenvalue spectra at the final checkpoint, MLP (`c_mlp`) against one copy of the SMCD circuit (`q_serial`), seed 0, top- $|\Omega|$ index range shaded. Left: Poisson. Right: Helmholtz $k=10$.

from about 4 to 12 on Poisson and 5 to 50 on heat. The size-matched classical model behaves like the circuit (effective rank 2.0 and 1.9), so this is an effect of size rather than of being quantum. Near-lazy dynamics are useful here: a model whose kernel barely moves is well described by its initial NTK, which is why the design card, computed at initialisation, predicts the trained behaviour.

Coverage alone does not predict error. Sweeping SMCD’s coverage target on Poisson builds only three distinct circuits (targets of 0.8, 0.9 and 1.0 yield the same one), so coverage there is confounded with parameter count; at the full training budget the correlation between coverage and error is $\rho = +0.29$, not the pre-registered $\rho \leq -0.7$. On Helmholtz $k=10$ every target builds the same circuit, which scores about 1 throughout. Covering the band is necessary for the design to help (Section 5.3) but not sufficient for a single circuit to train well.

No measurable barren plateau at this scale. Gradient variance fitted against qubit count (4 and 6 qubits, three seeds) has slope 0.31, under the $\log 2 = 0.69$ bar, but the fit explains almost none of the variation ($R^2 = 0.001$): no decay is detectable between 4 and 6 qubits. SMCD’s shallow depths and local observable are designed to avoid the plateau, and at this scale they do; eight qubits exceeded the simulator’s memory.

5.5 Why one copy is not enough, and what eight copies change

One copy learns the hard frequency and loses the easy one. The Poisson solution is $\sin \pi x + 0.3 \sin 15\pi x$. Tracking the error at those two frequencies (Figure 5), one copy cuts its 15π error to a median 10% of its starting value by step 1,000, faster than any classical model (Fourier features 103%, MLP 86%, frequency-matched 51%). Its π error falls to about 30% by step 5,000 and then *rises* to $1.9\text{--}2.0\times$ its starting value by step 20,000, on all five seeds, while the frequency-matched model reduces both.

A worked number explains the trade. The PDE residual contains u'' , and differentiating $\sin \omega x$ twice multiplies it by ω^2 , so an error at 15π costs $15^2 = 225$ times as much loss as the same error at π . A linear head on fixed sine features sets the two amplitudes independently, so it can reduce both. A single circuit cannot: all 81 of its Fourier amplitudes are functions of the same ten angles, read out through one expectation value. Moving the angles to fix the heavily weighted 15π amplitude moves the lightly weighted π amplitude too, and the optimiser accepts that, because the loss it saves at 15π is 225 times larger than what it loses at π . The loss agrees: it stalls near 3.3×10^4 from step 1,000 while the error keeps moving, and the seeds agree closely (worst over best $1.1\times$) because they all converge to the same trade-off. This reading is supported by the data but not proven.

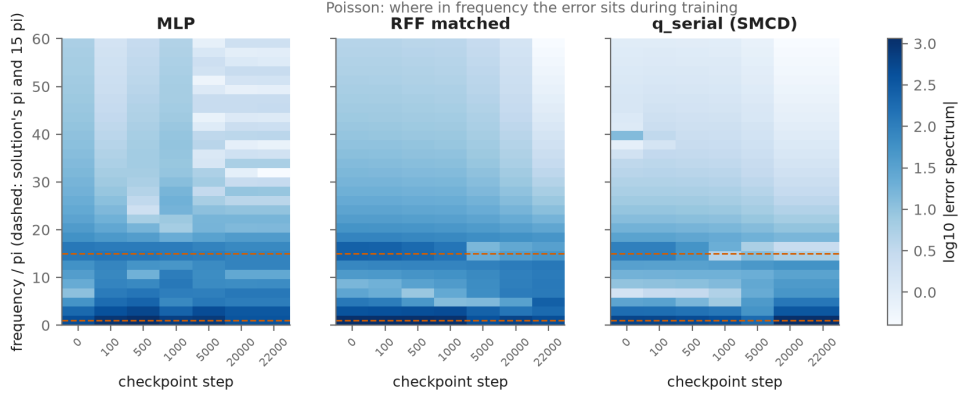


Figure 5: Poisson, one copy: error spectrum at each checkpoint (seed 0), with the error at π and 15π tracked over training.

Copies decouple the amplitudes. With K copies the head has K independent weights on K independently parametrised circuits, all with the same Ω . On the tuning seeds the error falls with the number of copies, from 1.67 with one copy to 0.003 with eight at the higher learning rate (Figure 6): 4 copies already give 0.08, and the drop from one copy to eight is almost three orders of magnitude.

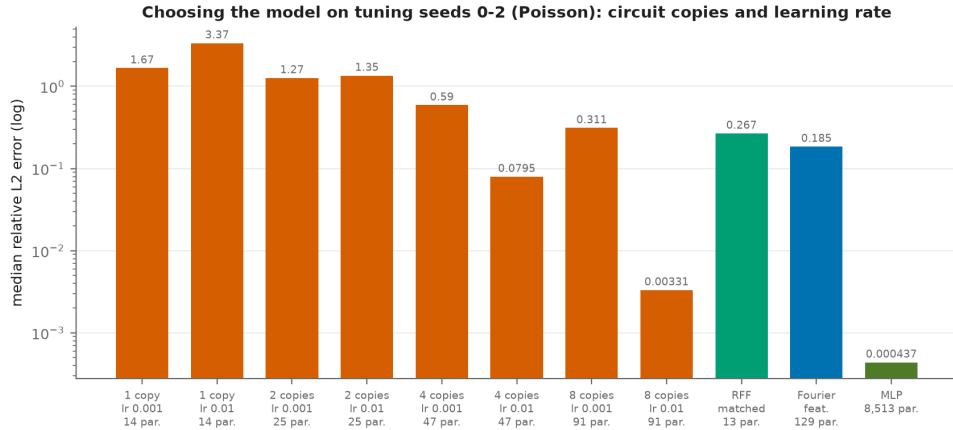


Figure 6: Model selection on Poisson, tuning seeds 0–2: median error for 1, 2, 4 and 8 copies at two learning rates, beside the best learning rate for each classical model.

Eight copies on held-out seeds. On seeds 10–14 (Table 4) the eight-copy circuit reaches a median error of 0.0037, with every seed below 0.02: from worse than predicting zero to about 0.4% error. It beats the Fourier-feature network by three orders of magnitude and the same circuit with random frequencies by 236 $\times$; the NTK is now flatter inside Ω (gap 15.2), and 99.9% of the improvement lies inside Ω . The comparison with the MLP is inconclusive: at its tuned learning rate the MLP failed on three of five seeds (1.2–3.6) and reached 0.003–0.012 on the other two. The classical learning rates chosen on the tuning seeds did not transfer well (the Fourier-feature network went from 0.19 to 5.0), which inflates both ratios, while every circuit seed was stable.

The frequency-matched classical model, sized to the eight-copy circuit (83 parameters), is 3.3 $\times$ more accurate (0.0011). On Poisson, handing a classical model the frequencies SMCD reads off the equation works better than the circuit; the contribution there is the frequency set, which a classical and a quantum model can both use. The groundwater problem (Section 7) is where the circuit’s own trainable encoding makes the difference.

Claim	Verdict	Measured
Beats the large MLP on Poisson ($\geq 2\times$)	Inconclusive	337.79 $\times$
Beats Fourier features on Poisson ($\geq 1.3\times$)	Confirmed	1359.81 $\times$
Matches the frequency-matched classical model on Poisson (within $1.3\times$)	Refuted	3.30 $\times$
Designed beats random frequencies, Poisson ($\geq 1.5\times$)	Confirmed	236.08 $\times$
NTK flatter inside Ω than outside, Poisson	Confirmed	gap 15.2
Gain over the MLP lies inside Ω , Poisson ($\geq 70\%$)	Confirmed	99.9%
Leaves the lazy-training regime earlier than classical models	Refuted	2.53 vs 9.7e+04
Encoder drift stays below 0.2, so Ω stays valid	Confirmed	0.10
Confirmed	5 of 8	

Table 4: Pre-registered claims about the eight-copy SMCD circuit on Poisson, held-out seeds 10–14. The baselines are sized to the eight-copy circuit where the claim names one.

6 Robustness of the verdicts

Every verdict comes from one pre-registered rule applied to five paired seeds. This section asks how much the verdicts on the six-problem matrix depend on that rule and on individual seeds. All of it is computed from the committed runs (`scripts/report/fig_validation.py`); no model was retrained.

Bootstrap intervals. Figure 7 resamples the five paired seeds 5,000 times and gives a 95% interval for each pre-registered error ratio. The heat design-versus-random result is robust: the designed circuit’s error is $6.5\times$ lower, with the whole interval ($2.5\times$ to $17.7\times$) above the $1.5\times$ bar. The one-copy Poisson comparisons against classical models fail decisively rather than narrowly: against the MLP the interval is 0.40 to 1.10 (bar $2\times$), against Fourier features 0.01 to 0.43, and against the frequency-matched model 3.9 to 5.8, entirely outside its equivalence band. The one-copy Poisson design-versus-random ratio (1.84, interval 0.97 to 7.2) straddles its threshold, as an inconclusive verdict should.

Other tests. With five paired seeds the smallest two-sided Wilcoxon p is $1/16 = 0.0625$ however large the effect, so the confirmation rule can only be met by five seeds of five agreeing. Three alternative tests on the same pairs agree with every verdict and show what the Wilcoxon floor hides: on the heat design-versus-random comparison a paired t -test on log error gives $p = 0.006$ and a Mann–Whitney test $p = 0.008$.

Leave one seed out. If one seed decided a verdict, dropping it would move the ratio across its threshold. With each seed removed in turn, the heat design-versus-random ratio clears its bar all five times, the one-copy Poisson comparisons against classical models miss theirs all five times, and the one-copy Poisson design-versus-random ratio clears it four times of five. No verdict rests on a single seed.

7 Application: waterlogging under canal-irrigated farmland

SMCD needs a solution’s frequencies before training. Real problems where those are known in advance come from regular structure: evenly spaced sources, periodic geometry. This section applies the method to one such problem, with its claims pre-registered like all others (Section 4.3).

7.1 The problem

Two rivers $L = 2$ km apart bound a confined aquifer beneath irrigated farmland. Water enters from background recharge R_0 (rain and irrigation return flow) and from six canals running parallel to the

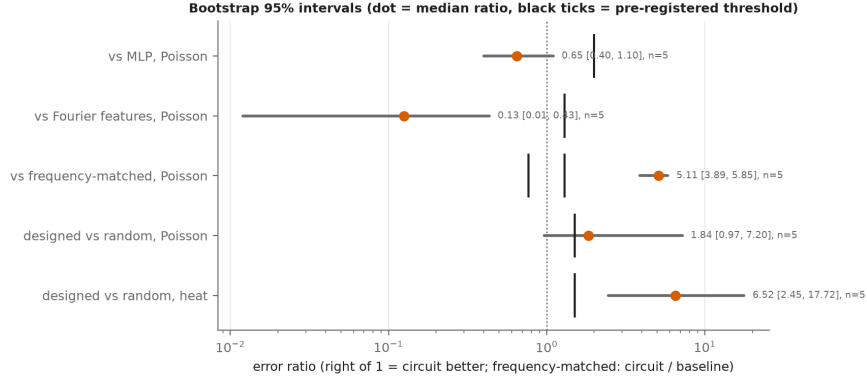


Figure 7: Bootstrap 95% intervals of the pre-registered error ratios, one copy of the circuit (5 paired seeds, 5,000 resamples). Right of 1 means the circuit is better; for the frequency-matched comparison the ratio is circuit over baseline and must lie between the two ticks.

rivers at equal spacing, each leaking q m³/day per metre of its length into a strip of width σ . Dupuit’s steady one-dimensional equation for the hydraulic head h (water-table height above the river datum) is

$$T h''(X) = -R(X), \quad R(X) = R_0 + \sum_{j=1}^6 \frac{q}{\sigma \sqrt{2\pi}} e^{-(X-c_j)^2/2\sigma^2}, \quad h(0) = h_A, \quad h(L) = h_B, \quad (1)$$

with $c_j = (j - \frac{1}{2})L/6$. The scenario uses values within textbook ranges [4, 13], not calibrated to one site: transmissivity $T = 100$ m²/day (silty sand), $R_0 = 0.4$ mm/day, $q = 0.4$ m²/day, $\sigma = 30$ m, $h_A = 1$ m, $h_B = 0$. On $x = X/L$ this is $-u'' = f$, the Poisson operator with non-zero boundary values, so every model and instrument applies unchanged. The canal spacing fixes the frequencies, harmonics of $2\pi \cdot 6$ in x , and SMCD designs a one-qubit, four-layer circuit for them.

Because $\sigma(z\Phi(z) + \phi(z))$ with $z = (X - c)/\sigma$ has second derivative $\phi(z)/\sigma$, the head has a closed form. An independent second-order finite-difference solver agrees with it to 5×10^{-8} m and converges to it at measured order 2.00 as the grid is refined from 100 to 6,400 intervals (Figure 9), which checks the closed form, the forcing and the boundary treatment together.

7.2 The practical question

A field is waterlogged where the water table rises within 1.5 m of the land surface, into the crop root zone. With the surface 9 m above the river datum, the water table peaks at 8.53 m and 713 m of the 2 km field is waterlogged (Figure 8). The solution answers the planning questions a farmer or irrigation engineer would ask (Figure 9):

- **Lining the canals.** Cutting canal seepage by 20% dries the whole field; a 10% cut reduces the waterlogged length from 713 m to 494 m.
- **Soil.** The same canals waterlog 1,778 m on clay loam ($T = 25$ m²/day) and none of the field on fine sand ($T \geq 150$).
- **Spacing.** Concentrating the same total seepage into two canals waterlogs 1,066 m and raises 84 cm mounds under each canal; spreading it over twelve reduces the mounds to 1 cm.

7.3 Can the trained models answer it?

Five claims were pre-registered for the SMCD circuit and tested on held-out seeds 10–14 at the full training budget, with one copy and with eight, each against baselines of its own size (Table 5, Figure 10).

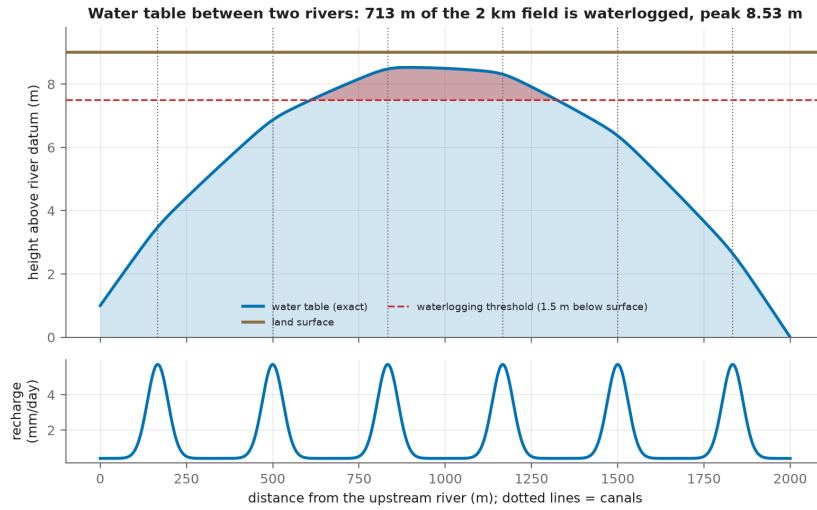


Figure 8: Exact water table between the two rivers (top) and the recharge that drives it (bottom). Red: the 713 m of field where the water table is within 1.5 m of the surface.

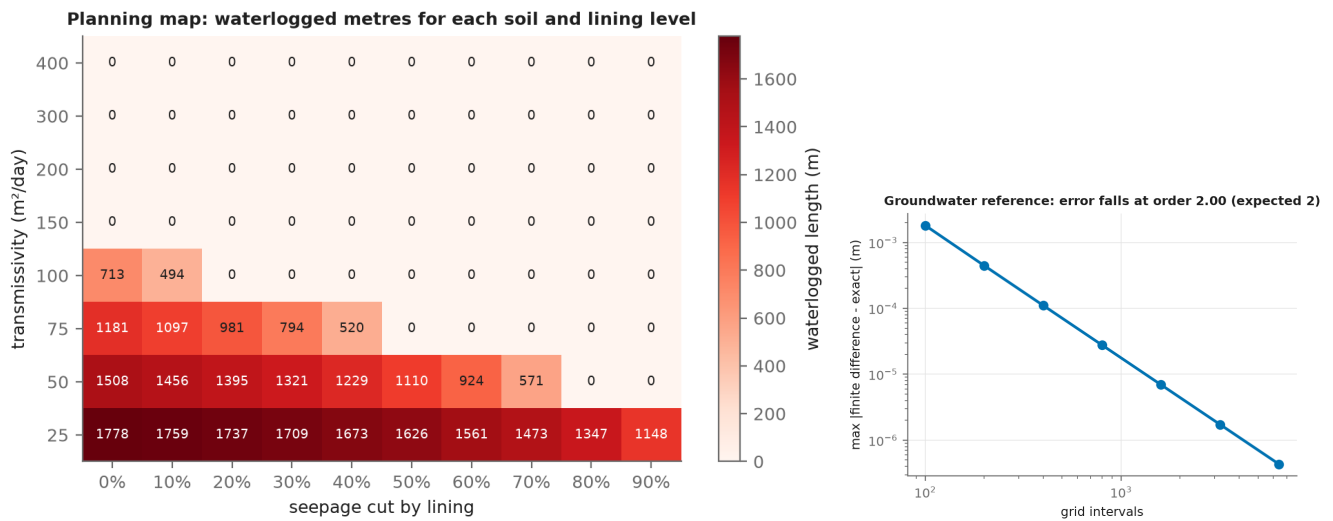


Figure 9: Left: waterlogged length for each combination of soil (transmissivity) and canal lining (seepage cut). Right: finite-difference error against the closed form as the grid refines.

Claim	1 copy	8 copies
Error below 1%	Refuted (52.76%)	Confirmed (0.09%)
Beats a classical network of the same size ($\geq 1.3\times$)	Refuted (0.09 $\times$)	Confirmed (18.23 $\times$)
Beats the same circuit with random frequencies ($\geq 1.5\times$)	Refuted (0.11 $\times$)	Confirmed (9.91 $\times$)
Gain over the MLP lies inside Ω ($\geq 70\%$)	Confirmed (99.4%)	Confirmed (99.4%)
Waterlogged length within 5% of the exact 713 m	Refuted (0 m)	Confirmed (714 m)
Confirmed	1 of 5	5 of 5

Model	parameters	median error	waterlogged (m)
SMCD circuit, 8 copies	91	0.09%	714
random frequencies, 8 copies	91	0.92%	738
MLP, size of 8 copies	91	1.68%	672
frequency-matched, size of 8 copies	83	22.93%	0
MLP, large	8,513	1.94%	707
SMCD circuit, 1 copy	14	52.76%	0
random frequencies, 1 copy	14	5.96%	535
MLP, size of 1 copy	13	4.96%	840
frequency-matched, size of 1 copy	13	15.76%	0

Table 5: Groundwater: pre-registered claims for the SMCD circuit with one and eight copies (top) and every model’s median relative L^2 error and waterlogged length (bottom), held-out seeds 10–14. Exact waterlogged length: 713 m.

Eight copies meet all five claims. The median error is 0.09%, with every seed below 0.5%. On a water table that peaks at 8.5 m, 0.09% is a typical error of a few millimetres, far below what any field measurement could resolve. The circuit is $18\times$ more accurate than an MLP with the same 91 parameters and $9.9\times$ more accurate than the same circuit with random frequencies, both on every paired seed, and it is also more accurate than the 8,513-parameter MLP, with about 1% of its parameters. 99.4% of its improvement over the MLP lies inside Ω , so the canal harmonics SMCD read off the forcing are where the accuracy comes from.

It answers the practical question. Relative error is not what a planner needs; the waterlogged length is. The eight-copy circuit places it at 714 m against the exact 713 m. The same-size MLP, with 1.7% error, gives 672 m (6% short, outside the 5% claim), because a small error near the 7.5 m threshold moves where the water table crosses it. Accuracy at the level of the planning decision requires accuracy well below 1%.

Here the circuit beats the frequency-matched classical model. Unlike on Poisson, the classical model handed the same frequencies fails here (22.9% error, no waterlogging predicted). A plausible reason, not tested here, is that the solution is a broad mound plus small canal ripples; a linear head on fixed sine features at the canal harmonics represents the ripples but not the mound, while the circuit’s trainable encoder can shift and scale its frequencies to fit both.

One copy meets one of five. Its improvement over the MLP still lies on the designed frequencies (99.4%), but its median profile peaks near 4 m instead of 8.5 m, so it predicts no waterlogging at all (Figure 10, right). This is the Poisson trade-off of Section 5.5 again: the broad mound is the slow component, and a single circuit gives it up to fit the canal ripple.

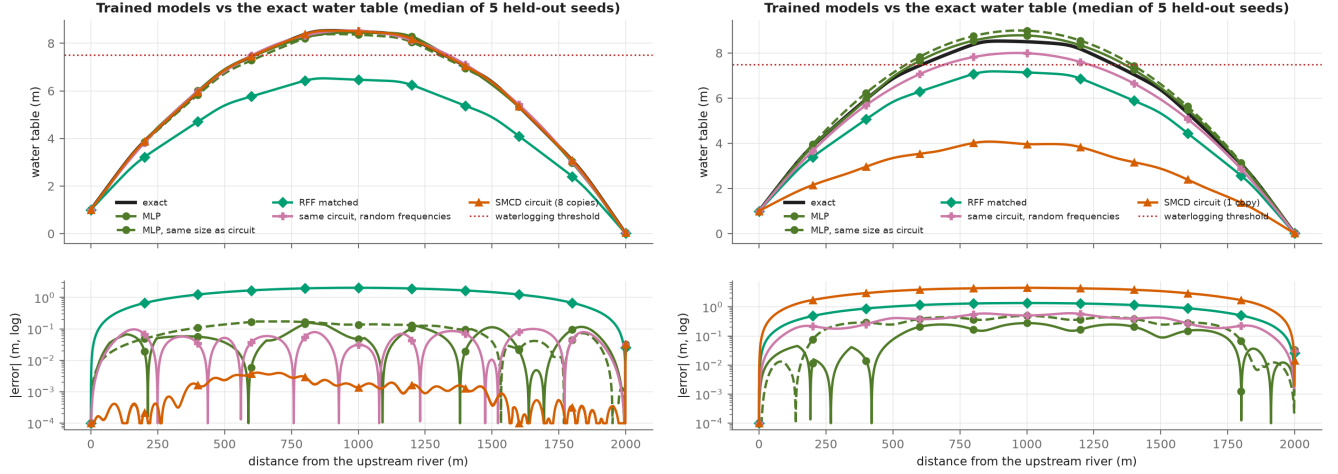


Figure 10: Predicted water table of each model against the exact solution (median of seeds 10–14), with the absolute error along the field below. Left: eight copies of the SMCD circuit and baselines of its size; the circuit stays within a few millimetres. Right: one copy and baselines of its size. Dotted: the waterlogging threshold.

8 Limitations

1. **Simulation only, small circuits.** Every circuit runs on a statevector simulator; nothing here ran on quantum hardware, and noise is modelled by analytic surrogates checked against a density-matrix simulation. On the one-dimensional problems each circuit is one qubit without entanglement, which a classical computer evaluates cheaply, and simulation cost capped the experiments at 6–8 qubits. No claim of quantum computational advantage is made or tested.
2. **Where the method wins.** The eight-copy circuit’s strongest result is one applied problem. On Poisson a classical model given the same frequencies remains $3.3\times$ more accurate, and at high Helmholtz wavenumbers no model learns the solution, so parity there says nothing about capability.
3. **Statistical power and baselines.** Five seeds give a smallest Wilcoxon p of 0.0625, so “confirmed” means all five seeds agree. In the six-problem matrix only the frequency-matched model and the random-frequency circuit are size-matched; the MLP and Fourier-feature network run at standard sizes. The eight-copy model was selected from a small grid of copy counts and learning rates, while classical families received a learning-rate search only.
4. **Design assumptions.** The encoder is restricted to an affine map so that Ω and coverage are well defined; achieved coverage does not by itself predict error; and the design card’s predicted benefit is a heuristic, not a bound (it predicted 2.5% on heat and zero on Helmholtz $k=4, 10$, and the circuit beat the MLP on none of them).

9 Conclusion

Spectrum-Matched Circuit Design turns the choice of a quantum circuit for a PDE into arithmetic: the operator’s Fourier symbol and the forcing give the frequencies the solution needs, and the encoding scalings, depth, qubit count and entangler follow by formula. The circuit it builds produces exactly the designed frequencies, and those frequencies matter: against the same circuit with random ones, the designed circuit is $6.5\times$ more accurate on the heat equation, $236\times$ on Poisson and $9.9\times$ on groundwater, on every seed.

The explainability instruments showed what a single circuit cannot do. All of its Fourier amplitudes depend on the same few angles, so on a solution that mixes a slow and a fast wave the optimiser trades one for the other. Mixing eight copies of the designed circuit through a linear head keeps the frequency set and frees the amplitudes. On waterlogging of farmland from leaking canals, that model reaches 0.09% error, $18\times$ more accurate than a classical network of the same size, and answers the planning question to within 1 m.

The results also mark the method’s limits. On a smoothing operator no model beats a plain MLP, as SMCD predicts before training; on Poisson a classical model handed the same frequencies is $3.3\times$ more accurate than the circuit; and at high Helmholtz wavenumbers every model fails. The most direct next steps are running the designed circuits on quantum hardware, extending SMCD to problems of higher dimension, and understanding when a trainable encoder lets the circuit beat fixed classical features at the same frequencies, as it does on the groundwater problem.

A Reproducibility

The code, every run’s metrics and instrument records, and the scripts that turn them into every figure, table and verdict are in the project repository; no figure reloads a trained model. All runs used one machine (NVIDIA RTX 4090 Laptop GPU, 32 CPU cores) with Python 3.12, PyTorch 2.11 (CUDA 12.8), PennyLane 0.45, NumPy 2.4 and SciPy 1.18, pinned by `uv`. A single integer seeds Python, NumPy and PyTorch with deterministic algorithms, so two runs at the same seed on the same device give bit-identical metrics. Seeds: six-problem matrix 0–4; coverage and barren-plateau sweeps 0–2; model selection for circuit copies 0–2; held-out evaluation of the eight-copy circuit and all groundwater models 10–14. `tasks.py test` runs the test suite (about 4 minutes on CPU), `tasks.py figures` redraws every figure from the committed runs (about 30 s), `tasks.py repro-quick` trains all problems and families at reduced size (about 9 minutes on CPU), and retraining all 327 recorded runs from scratch takes about 130 hours on the machine above.

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